Chiral constructions and regular subgroups in dimension three

Principal synthesis for Leemans–Vandenschrick

We recover the three principal theorems of *On chiral polytopes having a group* $\mathrm{PSL}(3,q)$ *as automorphism group*, in its fixed 25-page author edition. The constructions and their original matrices are due to Leemans and Vandenschrick. The accompanying papers give complete new proofs, distinguish their classical inputs, and record several corrections to intermediate formulas and tables. The final unitary examples in Section 7 are supplied with explicit generators. Earlier computational statements cited there remain attributed prior results, with their evidence boundary stated below.

The common ideas are the actual translation kernels of small projective subgroups, elementary matrix generation, and the action of an involution's centre and axis. A subgroup order does not by itself prove an intersection property: the constructions use the actual common subgroups, and the finite unitary examples check every indexed intersection. A separate centralizer calculation excludes reversing automorphisms of the whole ambient group.

The three principal conclusions

Let $q = p^f$, and let $d = \gcd(3, q-1)$. We use the standard string rotation correspondence for abstract regular and chiral polytopes.

Theorem 1. *The following assertions hold.*

1. *For every $q \geq 5$, the three matrices in Theorem 1.1 of the source, with a primitive element $x \in \mathbb{F}_q^\times$, generate a self-dual chiral polytope with automorphism group $\mathrm{PSL}_3(q)$ and type*

$$[q-1,\ 2(q-1)/d,\ q-1].$$

   The adjacent subgroups are the full set stabilizers of two distinct lines and two distinct points, respectively, of order $2q^2(q-1)^2/d$.

2. *For every prime $p \equiv 1 \pmod 6$, each of the four markings specified in Theorem 1.2 of the source gives a chiral polytope with automorphism group $\mathrm{PSL}_3(p)$. Their types are $[3,6,6,3]$ for the positive sign and $[3,6,3,3]$ for the negative sign. Including duals there are six distinct oriented marking classes in this construction, which give three unbased polytope classes after identifying the two opposite base-flag orbits. The same matrices over $\mathbb{F}_{p^2}$, with $p \equiv 5 \pmod 6$, give directly regular examples: $\mathrm{PSL}_3(p^2)$ is the rotation group and its extension by Frobenius is the full group.*

3. *For every $q \geq 2$, neither $\mathrm{PSL}_3(q)$ nor $\mathrm{PSU}_3(q)$ is the automorphism group of a chiral polytope of rank greater than five.*

Proof. The first accompanying paper, *Two stabilizers and elementary matrices in dimension three*, retains the literal source matrices and duality. Its finite-field span argument generates both full translation radicals, hence all six elementary root subgroups. This proves ambient generation without a maximal-subgroup classification. The stabilizer intersection gives the rotation intersection property. Its final semilinear calculation includes the inverse-transpose coset and proves chirality, while direct multiplication gives the type and self-duality. This proves the

first assertion for every prime power in the stated range, rather than only the fields used in supplementary controls.

A parabolic subgroup and a conic in the rank-five construction proves the exact translation module of the first adjacent subgroup and the restricted parabolic containing it. The two signs are treated separately. For the negative sign a literal symmetric-square representation gives the conic end; the positive sign uses the stated point-pair duality. Both actual middle intersections and every recursive rotation intersection are proved. The common centralizer of three prefix involutions is scalar, excluding all reversing automorphisms over the prime field. Frobenius reverses the quadratic-field markings. The comparison of both cube roots and both signs, including duals, gives exactly the oriented-class statement above. This proves the second assertion with its orientation and full regular-group qualifications.

For the third assertion, the paper *The two regular ends of a chiral marking* first derives the two directly regular ends from the actual indexed rotation intersection property. The acquired regular subgroup classification bounds their ranks. A rotation of order two gives a proper normal direct product, so it cannot occur in the relevant simple ambient groups. The elementary unitary argument for $\mathrm{PSU}_3(2)$ pays the nonsimple boundary separately. It remains to exclude rank six. In even characteristic the two modules and their actual centres and axes force a pair of distant rotations not to commute. In odd characteristic the Heisenberg normal form gives the same contradiction by an explicit commutator. The remaining directly regular simplex case has rotation group a quotient of A_7; group orders exclude it in all ambient fields of characteristic five. The complete argument is uniform and proves the claimed bound, including all its small-field premises. □

The auxiliary classifications and source tables

The source's Section 5 is recovered by four accompanying components. *Small regular subgroups and their translation kernels* proves the rank-two order classification and the complete odd-characteristic alternatives in rank three. *Elations and two translation modules* proves the even-characteristic normal forms, the exact field and intersection conditions, and the regular rank bound. *Four homologies in a projective plane* gives the odd rank-four classification, including its degenerate alternative. The conic branch reuses the already proved projective-line classification: the exceptional markings are those for $\mathrm{PSL}_2(11)$, $\mathrm{PSL}_2(19)$ and $\mathrm{PGL}_2(5)$. Their exact presentations and checks remain acquired inputs.

These proofs use Dickson's subgroup types and the classical three-dimensional conic representation as attributed inputs. They do not count those classifications as new results. In the odd affine branch, writing $K = \mathbb{F}_p(a)$ identifies the actual kernel as K^2; the complements are literal translates, so their intersections can be calculated before appealing to the string end criterion. In even characteristic the corresponding two modules determine precisely when the marking has the intersection property. This accounts for all group alternatives and types used in Tables 2–3. The earlier rank-three projective-line classifications cited by those tables remain attributed inputs, with the actual string and intersection hypotheses retained. No automatic passage from an arbitrary hypermap marking is used.

The separate calculation *The parameter restrictions in Table 1* pays all nine signed parameter rows of that auxiliary table. It computes the three characteristic polynomials and eliminates t from $t^2 = 5$ and the exact middle-order equations. The listed primes are necessary parameter restrictions; the actual square root must belong to the ambient field and satisfy the displayed equation. Neither the table nor its heading is used as an independent intersection certificate.

The source corrections have precise limits. The first elementary-matrix word on physical page 6 needs an x^{-1} factor in one coordinate; the generation paragraph on page 7 interchanges its end-subgroup names. In Section 4 the conic sign requires its own intersection argument; the adjacent translation kernel over a quadratic field has order p^2, and the regular analogue has an

additional Frobenius automorphism. The six-class sentence is qualified by the oriented-marking convention.

On page 11 a regular unipotent has order p in odd characteristic, not the printed bound $2(2,p)$. The final dihedral divisibility bound survives. On page 17 the direct product is D_{2p}^2, not D_{2p}; the final characteristic-polynomial coefficient is a', not b. The point/line statements are qualified by duality. The simplex alternative needed in Section 6 occurs in characteristic five, allowing embeddings into extension fields, rather than only the single ambient field of order five.

Finally, the generic affine row of Table 3 needs a type correction. If $\lambda+\lambda^{-1}=2a$, put $h=|\lambda|$ and $h'=|-\lambda|$. Its actual types are

$$[h,h'], \quad [h,2p], \quad [h',2p],$$

not just the printed $[k,4]$ and $[k,k]$. For example $p=5, a=2$ gives three C-markings in the group of order 300, with types $[3,6]$, $[3,10]$ and $[6,10]$. The uniform proof and full finite checks agree. This repairs a row without changing the principal constructions or the rank obstruction.

Section 7: positive examples and retained prior results

Three explicit unitary examples in rank four recovers the three positive $\mathrm{PSU}_3(17)$ examples reported in the source, with types $[3,7,3]$, $[3,8,3]$ and $[3,9,3]$. Its matrices are defined by three Hermitian centres over $\mathbb{F}_{17}[\beta]$, $\beta^2=3$. Fresh exact permutation calculations prove full ambient generation and all 768 indexed intersections. A written calculation of the full semilinear centralizer excludes every reversing automorphism. This is a positive existence statement; no absence of other types is inferred from the source's sentence or from the exploratory search.

The negative lists for $\mathrm{PSL}_3(q)$ with $q\le 4$ and for $\mathrm{PSU}_3(q)$ with $q\in\{2,3,4,5,7,9,13,16\}$ are explicitly attributed in the source to earlier computational work. The three $\mathrm{PSU}_3(11)$ pairs, of types $[5,4,5]$ twice and $[4,5,4]$ once, are likewise attributed there to data collected for Buekenhout, Leemans and Tranchida. We retain these as prior source-attributed results, not as our independently repeated censuses. The available edition of that earlier paper provides timings and group data but not those output tuples; the project's accepted algorithmic recovery does not manufacture the missing execution receipts. This evidence boundary is explicit in the retained-input register.

The same paragraph's negative list for $\mathrm{PSL}_2(q)$ contains one incorrect entry: $q=81$. The acquired projective-line theorem gives a chiral example of type $[40,40,40]$. Indeed its pointed construction applies with $p=3$, $d=4$ and $k=40$: k is even and divides $(81-1)/2$, while none of $3^e-1, 3^e+1$ for $1\le e<4$ is divisible by 40. This corrects the quoted list by an already proved theorem. The remaining entries of that list satisfy its negative criteria.

The previously known absence of chiral polyhedra for $\mathrm{PSL}_3(q)$ and $\mathrm{PSU}_3(q)$ is the theorem attributed to Breda d'Azevedo and Catalano. Together with Theorem 1(3), it gives the source's conclusion that a unitary chiral example can have only rank four or five. The other introductory classification statements are retained with their earlier authorship; the already acquired project results supply exact reuse links where available.

The historical question about unitary groups for $q>17$ is recorded without a new claim about its present status. The quoted composite-degree PSL_2 question is settled by the previously acquired coprime-subfield construction in the projective-line paper. Its full proof and earlier attribution are retained.